

Multivariate Time Series Forecasting of the Federal Funds Rate Using Economic and Market Indicators

Christina Salemme

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1 Objective

The current economic climate reflects significant uncertainty among both consumers and business owners. To address these challenges, the Federal Reserve plays a crucial role. In addition to maintaining financial and economic stability for the United States, the Fed's primary responsibility is to uphold its dual mandate: keeping inflation around 2% while promoting maximum employment. The Federal Reserve's primary monetary policy tool for managing economic growth, unemployment, and price stability is the federal funds rate, which serves as the foundational benchmark for short-term interest rates through the United States financial system. The Fed's Federal Open Market Committee (FOMC) sets a target range for this rate and implements additional monetary policy operations to maintain the effective rate within that range, making it one of the most closely watched and economically significant interest rates in the global financial system. Following the September FOMC meeting, the target range for the federal funds rate is set at 4.00% to 4.25%, marking the first adjustment since the rapid series of adjustments made at the end of 2024. The federal funds rate strategically stretches across multiple domains, from anchoring the entirety of the yield curve with longer-term Treasury yields, mortgage rates, corporate borrowing costs, and consumer loan rates all adjusting in response to target changes to influencing asset prices, equities, fixed income, currencies, and commodities. Even the broader scope of the economy is affected by rate changes, including credit availability, borrowing costs for businesses and consumers, asset valuations shifting wealth effects, and exchange rate movements influencing international trade competitiveness.

While the federal funds rate is a crucial metric for the Fed, the effective federal funds rate (EFFR), which is recorded daily, provides a more precise measure of how the target range is performing in practice. The effective federal funds rate (EFFR) represents the volume-weighted median of overnight federal funds transactions reported by domestic depository institutions, providing an accurate measure of actual market conditions rather than simply the Fed's target. This distinction is crucial because while the FOMC sets policy targets, the actual effective rate can deviate based on money market supply and demand dynamics, banking system liquidity conditions, and financial market stress. The rate is measured as an annualized percentage yield (typically ranging from 0% to 7% in the modern era) and is published daily by the Federal Reserve Bank of New York, creating a high-frequency time series ideal for detailed temporal analysis and forecasting.

The primary objective of this project is to accurately forecast the Effective Federal Funds Rate (EFFR) by analyzing macroeconomic and market data from early 2000 through the fourth quarter of 2025. The research aims to identify robust relationships between the federal funds rate and its economic and financial determinants across multiple business cycles, financial crises, and varying monetary policy regimes.

2 Outcome

The final outcome of this capstone project is the development of a robust multivariate time series forecasting model designed to predict the Effective Federal Funds Rate (EFFR). This project will also employ a machine learning model and then compare the accuracy between a time series forecasting model and an MLM for efficiently predicting the EFFR. The project will deliver the following:

- **Predictive Model:** A statistical framework that utilizes sixteen numerical independent variables—including Treasury yields, equity indices, and market volatility—to forecast future rate movements
- **Statistical Analysis:** A comprehensive evaluation of the dataset’s non-stationarity, supported by Augmented Dickey-Fuller (ADF) tests, rolling mean/variance plots, and Autocorrelation Function (ACF) structures.
- **Feature Importance Evaluation:** An analysis of which economic indicators, such as the 2-Year Treasury yield or the VIX "fear gauge," hold the most predictive power over Federal Reserve policy dynamics.
- **SARIMA Model:** A seasonal univariate model to establish a baseline forecast and account for any potential periodic effects such as FOMC meeting schedules.
- **Machine Learning Model (MLM):** A multivariate model (such as a Random Forest or Gradient Boosting Regressor) to capture non-linear relationships between the EFFR and market indicators.
- **Comparative Performance Dashboard:** A final analysis comparing the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) of the SARIMA and MLM models across different interest rate regimes.

3 Impact

This project benefits financial practitioners, researchers, and policymakers by providing a data-driven measure of actual market conditions versus mere policy targets. Given that the federal funds rate anchors the yield curve and influences mortgage rates, corporate borrowing, and international trade competitiveness, an accurate forecasting model assists in navigating global financial system uncertainties

4 Literature Review (25 points)

Below are the five papers that have conducted

Deep Learning in EFFR Forecasting: [2] explores the application of Deep Learning models for the Effective Federal Funds Rate, providing a modern baseline for comparing neural network performance against traditional econometric models. Alao utilizes five different algorithms—Random Forest (RF), LightGBM, Long Short-Term Memory (LSTM), Bilateral LSTM, and Gated Recurrent Unit (GRU)—against an ARIMA model.

While Alao found that these algorithms generally outperform ARIMA, only one test was statistically significant enough to indicate the absolute superiority of deep learning. My research departs from this approach by specifically honing in on the most valuable features to predict the EFFR. While the Alao study utilized over 500 variables, I intend to start with a baseline of the most necessary variables—such as treasury yields and labor market indicators—and build from there. This minimalist approach, combined with feature selection techniques, is designed to specifically prevent **overfitting** and determine if high accuracy can be maintained with a more streamlined data requirement.

Forecasting the Daily Federal Funds Rate: [6] compares two alternative one-day ahead forecasts: a simple random walk and an ARIMA model. This provides a foundational baseline for utilizing ARIMA-based methodologies in daily rate forecasting. This research establishes the baseline that ARIMAs are necessary over a simple random walk as they retain memory. The random walk forecast generally assumes that the next value in a time series will be the same as the last observed value, making it extremely naive. The ARIMA is Autoregressive (AR) to use past values of the time series (lags) to predict the current value, Integrated (I) which involves difference data to make nonstationary data stationary, and Moving Average (MA) which uses past forecasting errors known as shocks to adjust predictions and account for random disturbances. My project will build on this by testing if modern Machine Learning Models (MLMs) can provide a superior "nowcast" compared to these traditional statistical benchmarks.

Reviewing Federal Reserve Bank Reserve Policies: [4] examines the EFR via time series to evaluate the effectiveness of the Fed’s banking policies and observe patterns over time. By using both ARIMA and MLs to forecast, this work validates the dual-method approach I am adopting, particularly in observing how policy shifts manifest as structural breaks in the data. This working paper is extremely well written and provides a plethora of accessible information that I wish to emulate in my research paper and final presentation.

Forecasting Federal Funds Rates with AI: [3] introduces a hybrid framework using LSTM and GRU architectures combined with Large Language Model (LLM) sentiment analysis of FOMC statements.

This study is highly relevant to my proposal as it confirms that deep learning models are superior to ARIMA in handling policy “regime changes.” While the AMRO paper emphasizes the importance of qualitative data from LLMs, my approach will focus on optimizing the numerical feature selection (such as Treasury yields and labor indicators) to achieve high predictive accuracy without the high computational cost of a full LLM integration. This will allow me to test if the numerical “baseline” I am proposing can achieve comparable results to their high-parameter models.

Financial Time Series Forecasting - A Machine Learning Approach: [5] This paper examines the stock market and explores the efficacy of machine learning versus statistical baseline regression models for predicting volatile asset prices. By testing across various periodicities (daily, weekly, monthly), the authors established an 81% accuracy rate for directional trends. The authors divide the predictions into three forms “one of which is a classification problem, and the other two being a regression problem with a continuous output.” They seek to find the best features to give them an idea on the effectiveness of each variable for short term and long term prediction. The authors state that “the approach of evaluating multiple classifiers utilising the same dataset, we minimise the risk of focusing on a single classifier which may not be the best fit for the data at hand.” The two regression tasks are to forecast the next period price change (whether it goes up or down) and the next period price which is a continuous value.

This study is particularly relevant to my objective of forecasting the EFR because it provides empirical evidence on the risks of high-dimensional data. Dingli and Fournier observed that model performance deteriorated due to **overfitting** when the number of features exceeded 30. This directly supports my methodological decision to start with a baseline of critical variables—such as the 2-Year Treasury yield and the VIX—to ensure model robustness. Furthermore, their success in using both classification and regression targets provides a framework for my own comparative analysis between SARIMA and multivariate MLs like Random Forest.

Empirical Comparison of ML Models: [1] conducts a comprehensive large-scale evaluation of eight machine learning architectures for time series forecasting, including SVR, MLP, and Gaussian Process Regression. This research provides insight on model choice and gives me a set expectation for which models will perform the best

This research is highly relevant to my proposal as it establishes the **Multi-Layer Perceptron (MLP)**—a core component of my planned multivariate analysis—as a top-performing model for diverse time series data. Furthermore, the authors’ findings regarding the varied impact of **preprocessing methods** align with my identified need to address non-stationarity in the Federal Funds Rate (EFR) through specific transformations. This study provides the empirical justification for prioritizing MLP and neural network architectures as robust alternatives to traditional econometric baselines.

Comparative ML with Sentiment and Leadership Features: [7] advances federal funds rate forecasting by augmenting standard macroeconomic datasets with **sentiment analysis** of Fed communications and **leadership characteristics** of FOMC members.

This research is highly relevant to my proposal as it highlights the “human element” in policy rate determination. While my primary focus remains on a baseline of numerical indicators like Treasury yields and the VIX, this paper provides a valuable perspective on how **regime-dependent behavior**—which I identified in my preliminary analysis—can be partially explained by qualitative shifts in Fed leadership. This means that the inclusion of consumer sentiment and Fed Chair identity (Currently

Jerome Powerll) has improved forecasting accuracy. It justifies the inclusion of high-frequency market data as a proxy for the sentiment and expectations that these authors capture through more complex NLP techniques. The authors also concluding that Elastic Net, which is a regularized linear model, outperformed tree-based models, something important to note when I conduct model comparisons when examining the AI models sufficient for prediction.

In my opinion, the most important conclusion they drew is this: "One of the biggest lessons from this project was realizing that simpler models are often more powerful when the data supports them. While machine learning is exciting and valuable, it's not a magic fix-and if the underlying patterns are mostly linear, traditional approaches like Elastic Net still shine. Future work could leverage real-time natural language processing and deep learning to further refine forecasting frameworks." This supports that older models are still the go to, however, I believe this can be expanded upon through my project proposal. Comparing a plethora of methodologies in various dimensions (time series, MLMs, and traditional approaches) can provide the insight that policymakers and economists may find compelling for future rate change decisions.

5 Novelty

The novelty of this capstone lies in its multidimensional approach, capturing data across four distinct domains: the Treasury yield curve, equity indicators, commodity pressures (Crude Oil), and financial stress (VIX). Unlike simpler models, this research accounts for the "zero lower bound" period (2008–2015) and the unprecedented rapid tightening cycle of 2022–2024, utilizing high-frequency daily data to capture regime-dependent behavior. I understand that this is an ambitious task and that the Federal Reserve operates with many software engineers and data scientists working together to improve models, but there has been little research besides the first piece of literature solely focusing on comparing MLMs to ARIMA. Using machine learning methods for prediction in the financial and econometric prediction is not something new, but most existing research looks at exchange rates, treasury yields and stock market pricing.

6 Data Source(s) (10 point)

Currently this project will utilize two main data sources. First, the Federal Reserve Economic Data (FRED), maintained by the Federal Reserve Bank of St. Louis and what is considered to be the gold standard for U.S. macroeconomic and financial time series data. Second, Yahoo Finance, a widely-used financial data platform that provides market data including indices, volatility measures and commodities. Both sources are freely accessible to researchers and practitioners, ensuring replicability of results and enabling continued monitoring and updating of forecasting models beyond the initial study period. This data will be extracted in python by using the necessary packages (fredapi and yfinance) and codes for each variable, thus I cannot finalize any citations. The analysis seeks to forecast the Effective Federal Funds Rate (EFFR) as the primary dependent variable. Currently, the analysis incorporates seventeen numerical independent variables with over 6,250 observations, carefully selected to represent distinct dimensions of economic conditions, market expectations, and financial stability that economic theory and empirical evidence suggest influence Federal Reserve policy decisions and federal funds rate dynamics. The list is as follows

- Three-Month Treasury Bill Rate
- Six-Month Treasury Bill Rate
- Two-Year Treasury Yield
- Five-Year Treasury Yield
- Ten-Year Treasury Yield
- Thirty-Year Treasury Yield
- S&P 500 Index (For this index, the Open, Close, High, Low, and Volume are represented as individual variables)

- VIX Index (this index has the same number of variables as the S&P Index does)
- Crude Oil Price
- **Potentially** the price of gold

From previous economic insight, this project should also include some labor market indicators as well as important inflation metrics. These additions would well align with the Fed's dual mandate to achieve maximum employment and stable pricing (inflation and unemployment ideally at the 2% over the long run). This means that the data must be extracted for CPI and PPI as well as other price indices from the Bureau of Labor Statistics to include inflation, as well as employment. The one issue with this is that these indices are not daily, they are monthly and that does not necessarily fit with the EFFT, so some data engineering will be required to upsample the monthly data to daily or, the multivariate models used will align frequencies by incorporating calendar features. (The ARIMA cannot handle this but the MLMs can). Being that it is known that inflation expectations are crucial for monetary policy decisions, the University of Michigan Survey of Consumers Inflation Expectation will also be used. Money growth (M2) will also be added to the dataset as it has been a notably good predictor for monetary policymaking. Alas, more research must be done on the initial data variables that should be used.

7 Approach

1. **Data Collection and Pre-processing:** Aggregating FRED and Yahoo Finance data into a unified temporal dataset. Exploring other data sources, such as UMich's Survey of Consumers, as well as the BLS and BEA.
2. **Stationarity Testing:** Conducting Augmented Dickey-Fuller (ADF) tests and rolling statistics analysis to confirm the presence of unit roots
3. **Correlation Analysis:** Utilizing Autocorrelation Function (ACF) plots to determine time-dependence patterns and lag structures.
4. **Feature Engineering and Selection:** Addressing non-stationarity by transforming data (e.g., differencing) to move beyond "levels" analysis. Utilizing various classifying methods to weed out unfit features and comparing using RMSE and MSE for model usage
5. **Model Development:** Applying multivariate time series forecasting methods (e.g., VAR, VECM, AND Machine Learning approaches) to predict the EFFT.

8 Timeline

- **Weeks 1-2:** Finalize data collection, cleaning and stationarity transformations.
- **Weeks 3-6:** Feature and model selection, training, and hyperparameter tuning.
- **Weeks 7-8:** Evaluation of forecasting accuracy and impact analysis.
- **Weeks 9 - 11:** Final report and dashboard/UI finalization.

9 Possible Challenges

When tackling a time series analysis there are always issues that can come up. The first being non-stationarity. The EFFT exhibits extreme level shifts and regime-dependent variance that violate standard modeling assumptions, meaning that there must be some logarithmic transformations to conduct to make the data stationary. The second being data gaps, specifically with difficulty in sourcing gold pricing. I will investigate alternative sources or use Crude Oil as a proxy for commodity-driven inflation. I will have to conduct a stationarity analysis as well as plot autocorrelation functions to rule out lag persistence prior to continuing with a SARIMA and then eventually an MLM model.

References

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